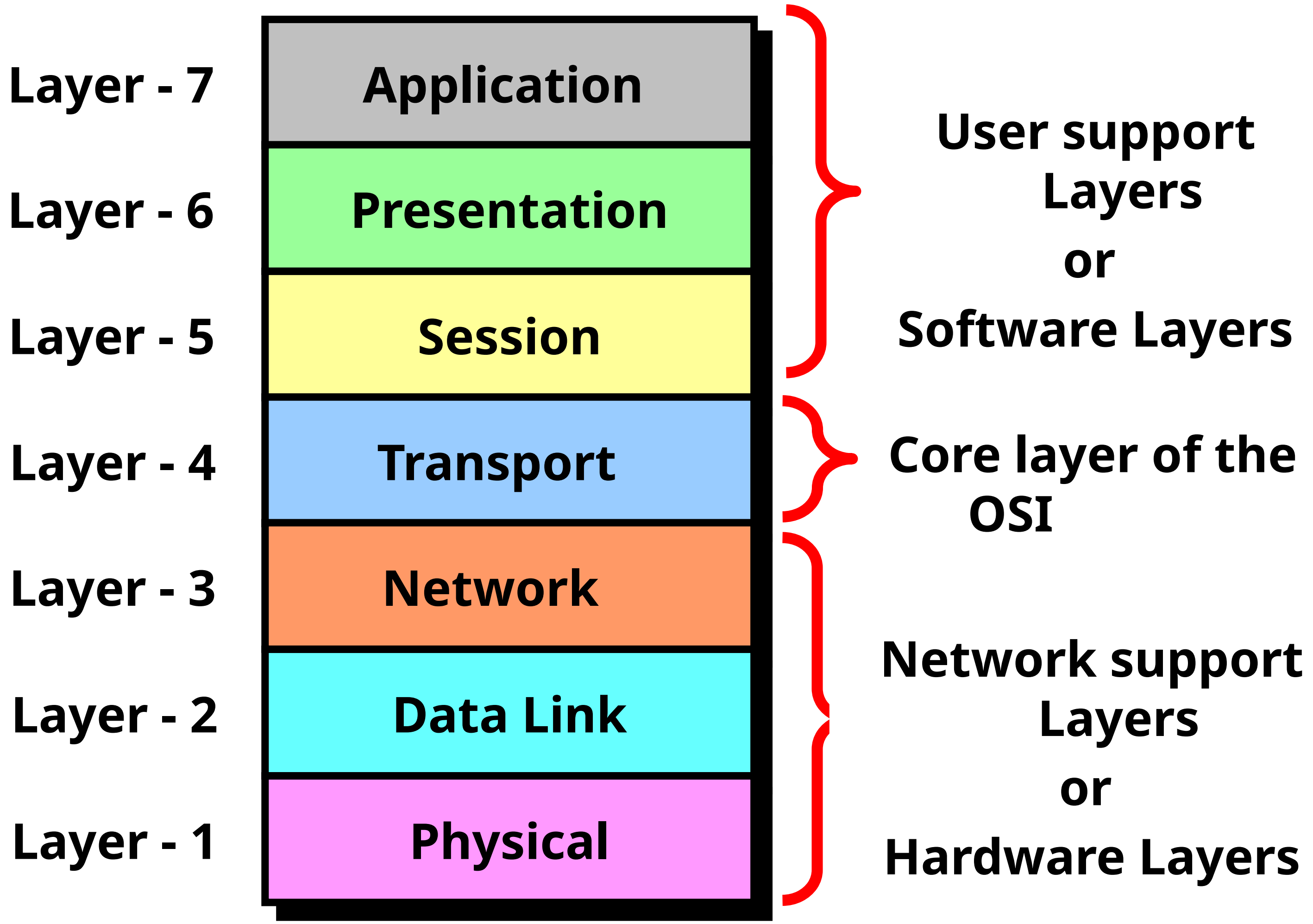
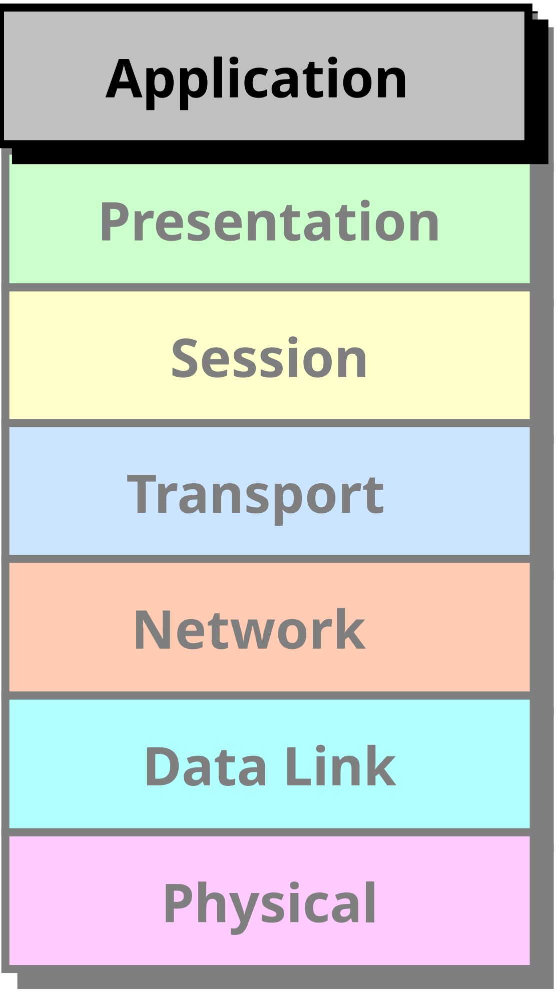
OSI (Open Systems Interconnect) Reference Model

- **OSI was developed by the International Organization for Standardization (ISO) and introduced in 1984.**
- **It is a layered architecture (consists of seven layers).**
- **Each layer defines a set of functions which takes part in data communication.**



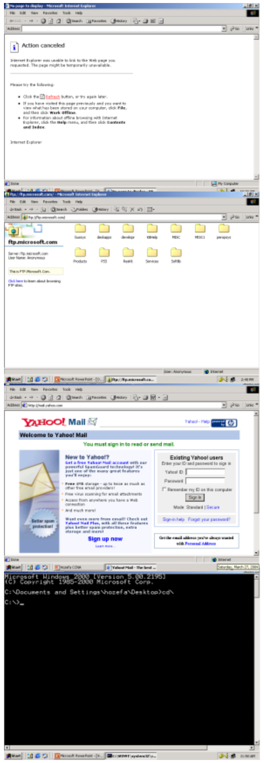
Layer - 7
Application
Layer - 6
Presentation
Layer - 5
Session
Layer - 4
Transport
Layer - 3
Network
Layer - 2
Data Link
Layer - 1
Physical
User support Layers or Software Layers
Core layer of the OSI
Network support Layers or Hardware Layers

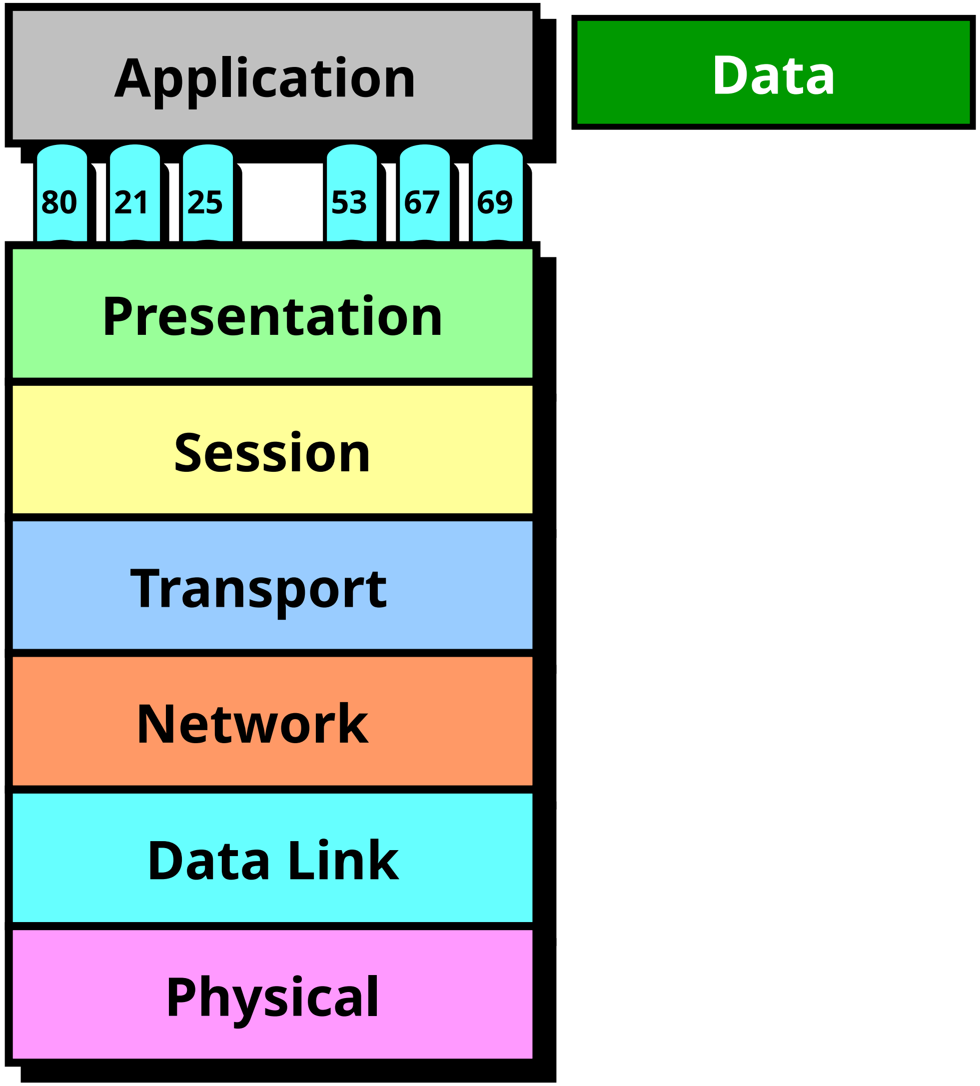


Application Layer is responsible for providing an interface for the users to interact with application services or Networking Services .

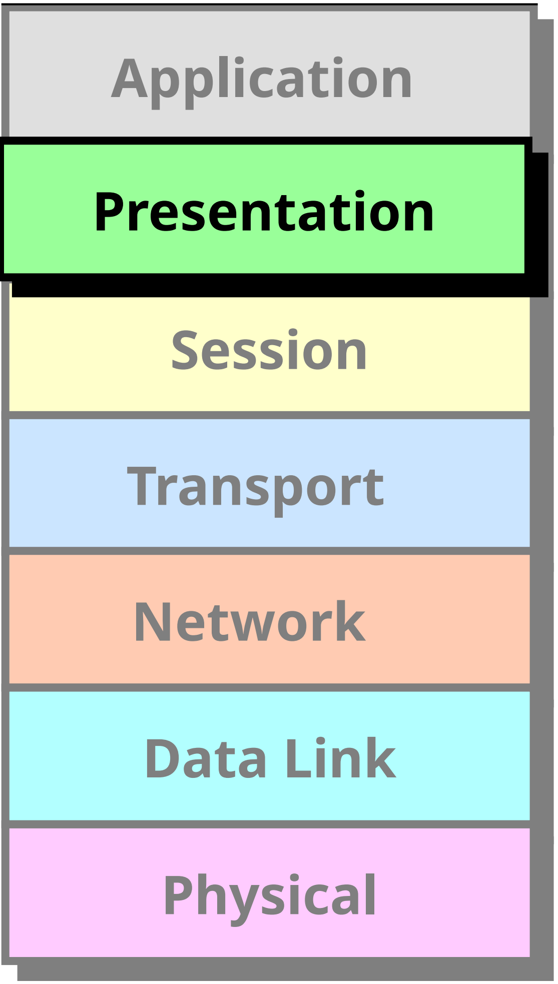
Ex: Web browser, Telnet etc.

Service	Port No.
HTTP	80
FTP	21
SMTP	25
TELNET	23
TFTP	69





Application
Data
80
21
25
53
67
69
Presentation
Session
Transport
Network
Data Link
Physical



Presentation Layer **It is responsible for defining a standard format to the data.**

It deals with data presentation.

The major functions described at this layer are..

Encoding – Decoding

Ex: ASCII, EBCDIC (Text)

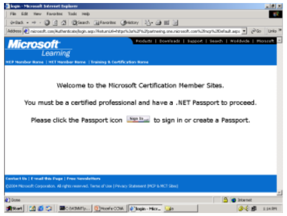
JPEG,GIF,TIFF (Graphics)

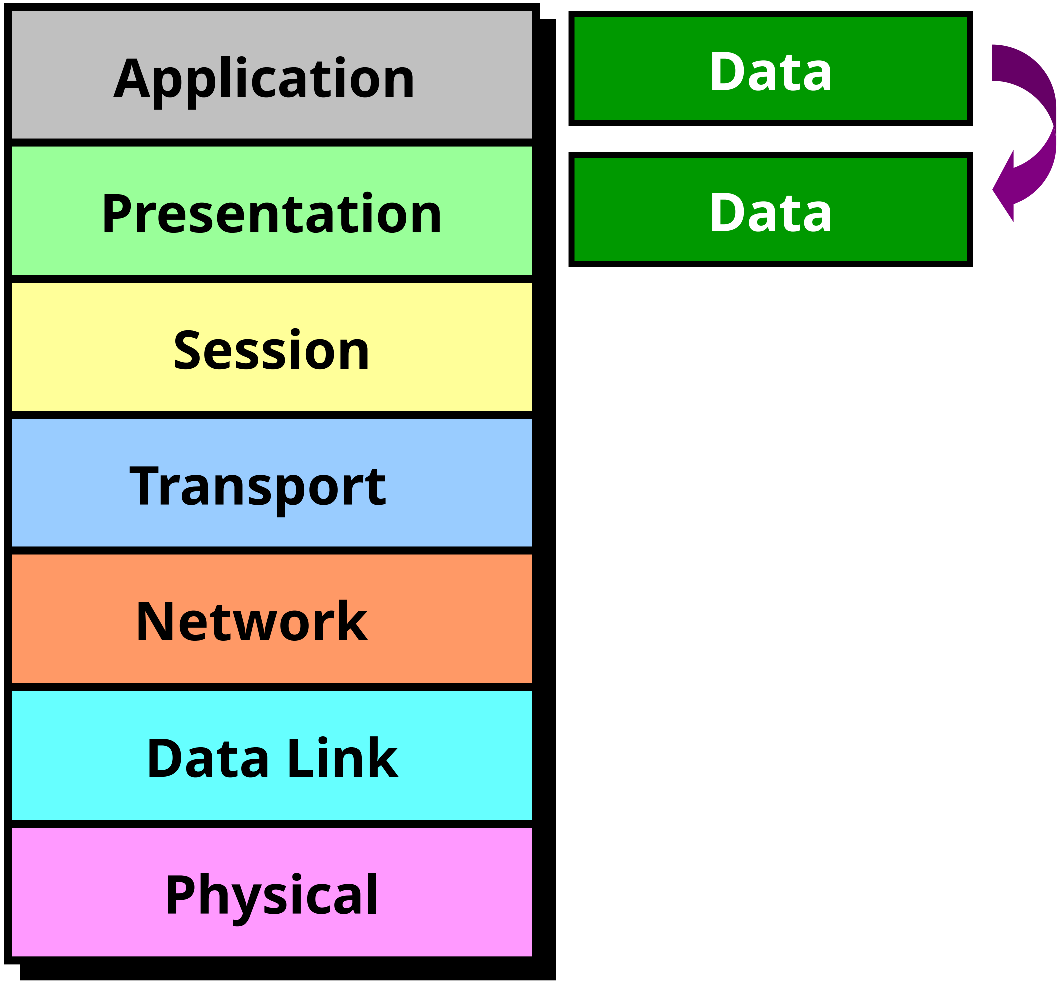
MIDI,WAV (Voice)

MPEG,DAT,AVI (Video)

Encryption – Decryption

Compression – Decompression




Application
Presentation
Session
Transport
Network
Data Link
Physical
Data
Data

Application

Presentation

Session

Transport

Network

Data Link

Physical

Session Layer

It is responsible for establishing, maintaining and terminating the sessions.

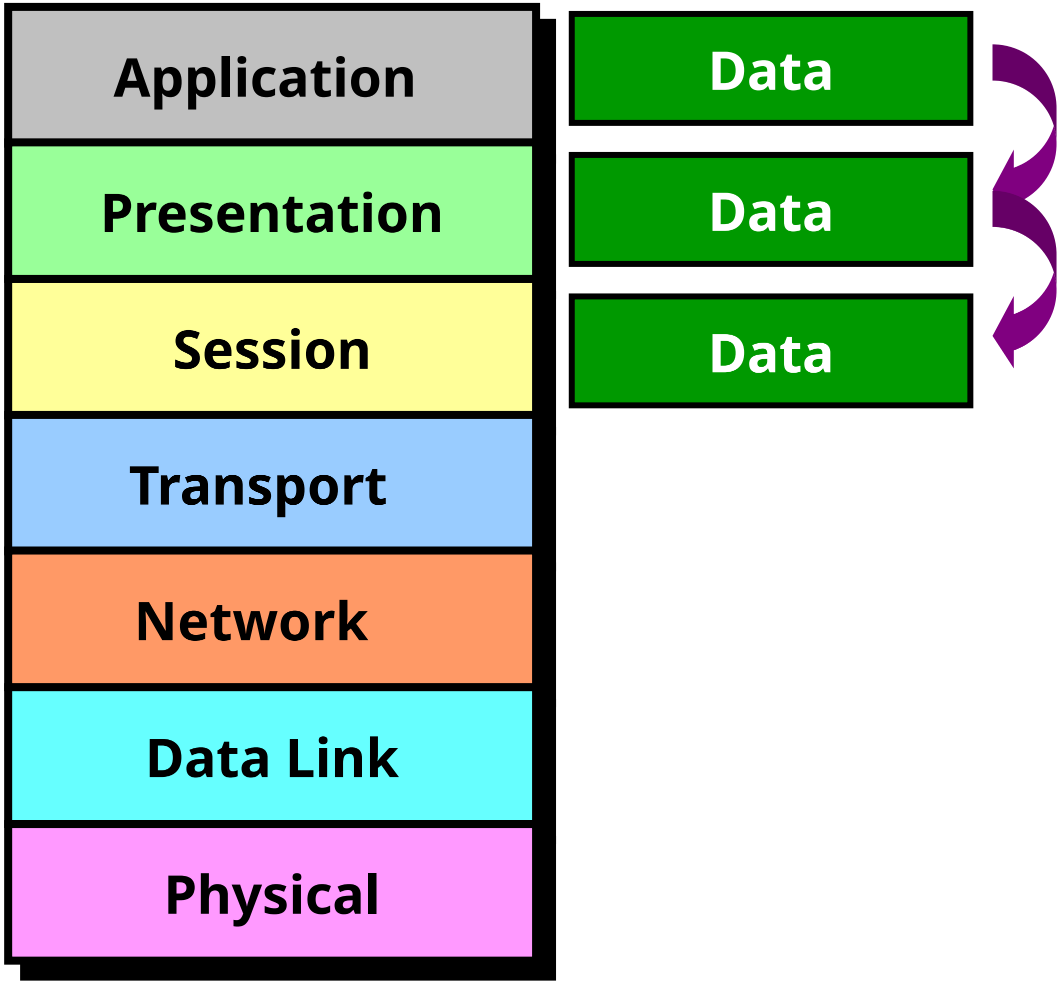
Session ID is used to identify a session or interaction.

Examples :

RPC → Remote Procedural Call

SQL → Structured Query Language

ASP → AppleTalk Session protocol


Application
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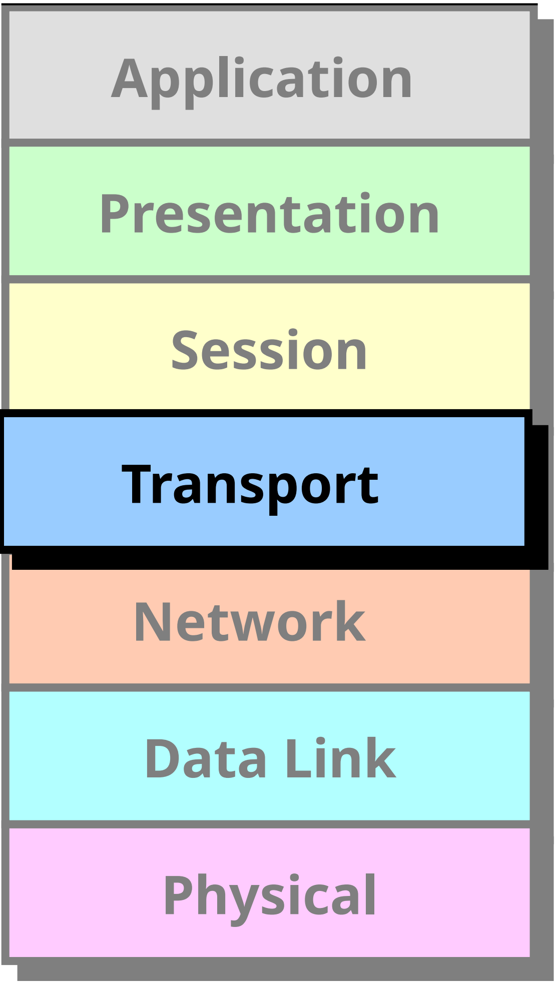
Transport
Network
Data Link
Physical
Data
Data
Data



Transport Layer

It provides data delivery mechanism between the applications in the network.

The major functions described at the Transport Layer are..

- **Identifying Service**
- **Multiplexing & De-multiplexing**
- **Segmentation**
- **Sequencing & Reassembling**
- **Error Correction**
- **Flow Control**

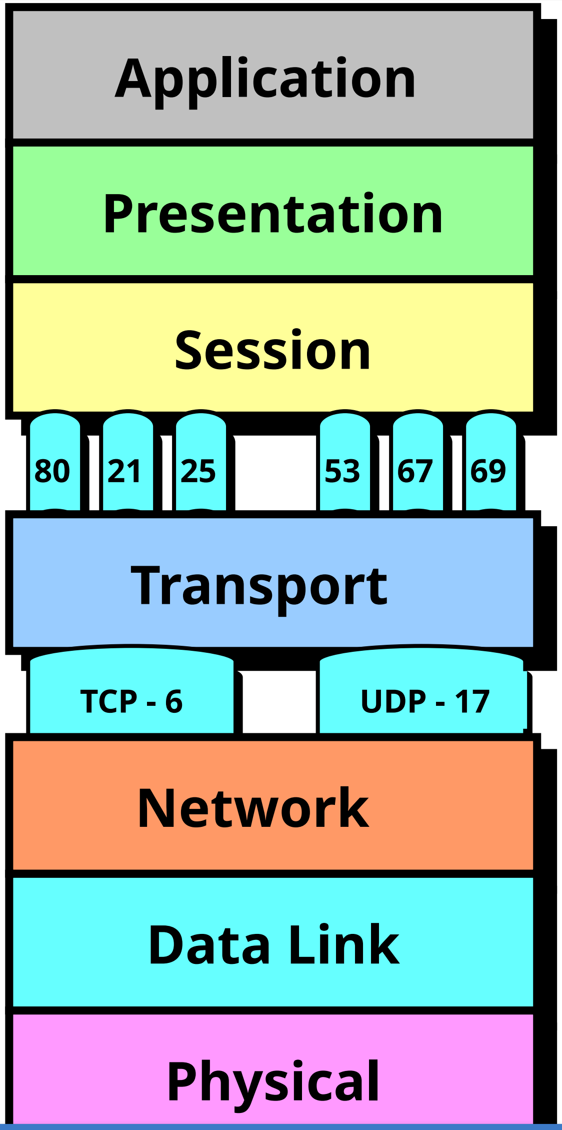
- **Identification of Services is done using Port Numbers.**
- **Port is a logical communication Channel**

Total No. Ports	**0 – 65535**
Server Ports	**1 - 1023**
Client Ports	**1024 – 65535**

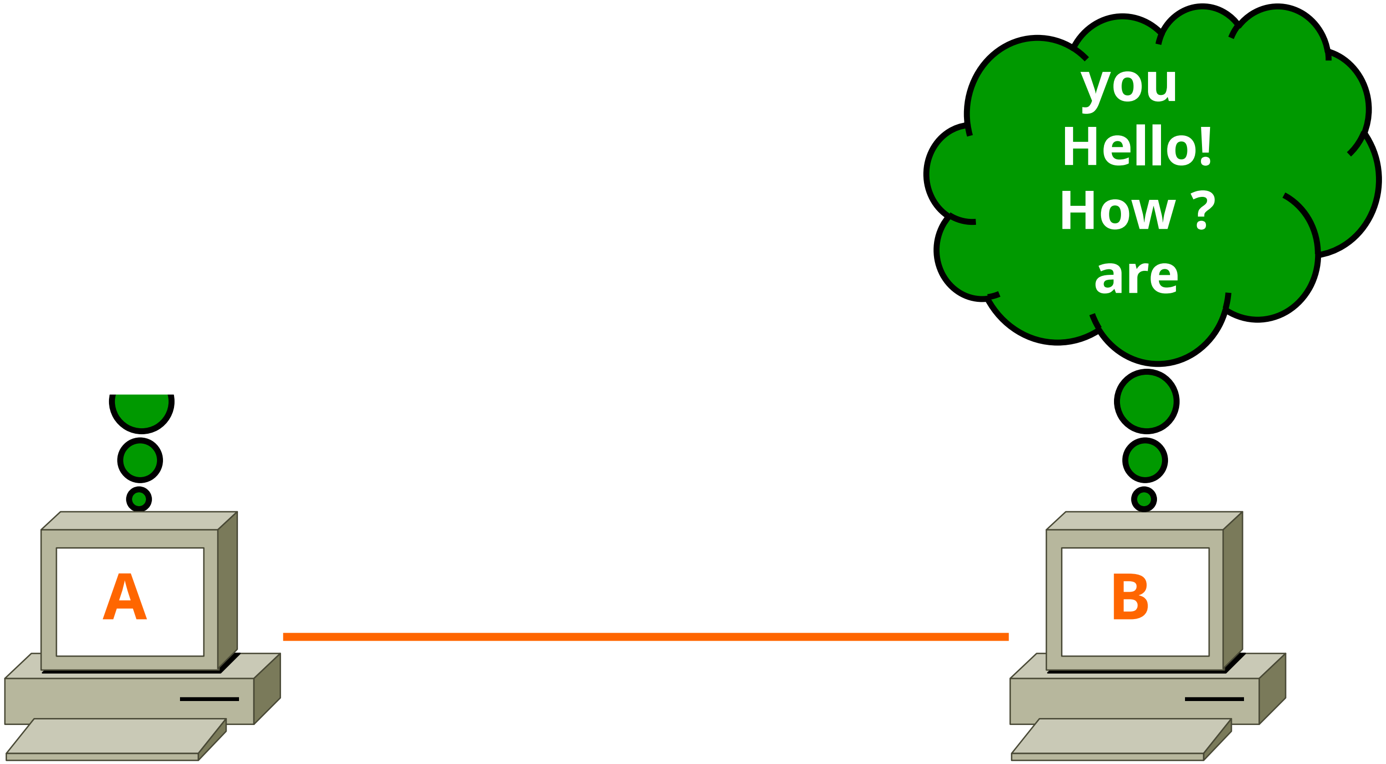
- **The protocols which takes care of Data Transportation at Transport layer are...TCP,UDP**

TCP	UDP
• Transmission Control Protocol	• User Datagram Protocol
• Connection Oriented	• Connection Less
• Supports Ack's	• No support for Ack's
• Reliable communication	• Unreliable communication
• Slower data Transportation	• Faster data Transportation
• Protocol No is 6	• Protocol No is 17
Eg: HTTP, FTP, SMTP	Eg: DNS, DHCP, TFTP

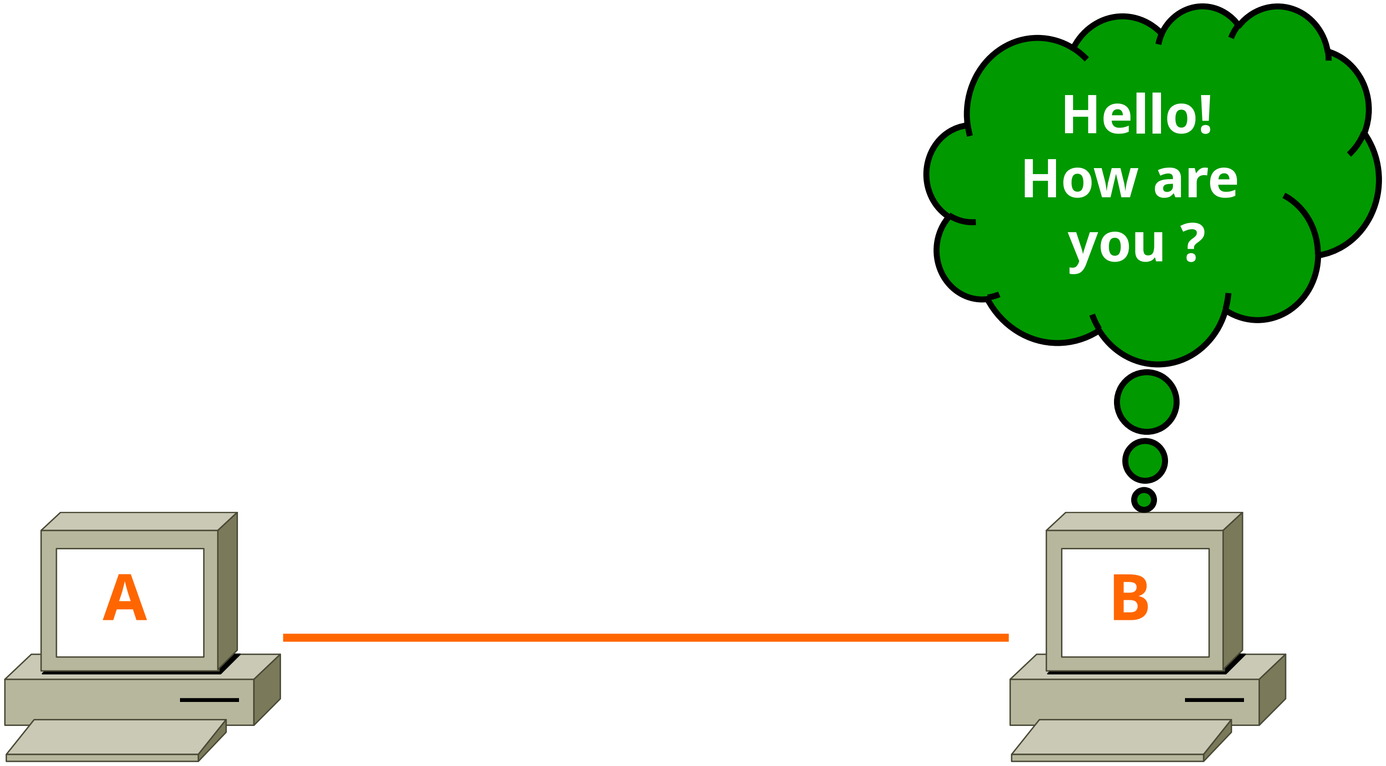
Multiplexing & De-multiplexing



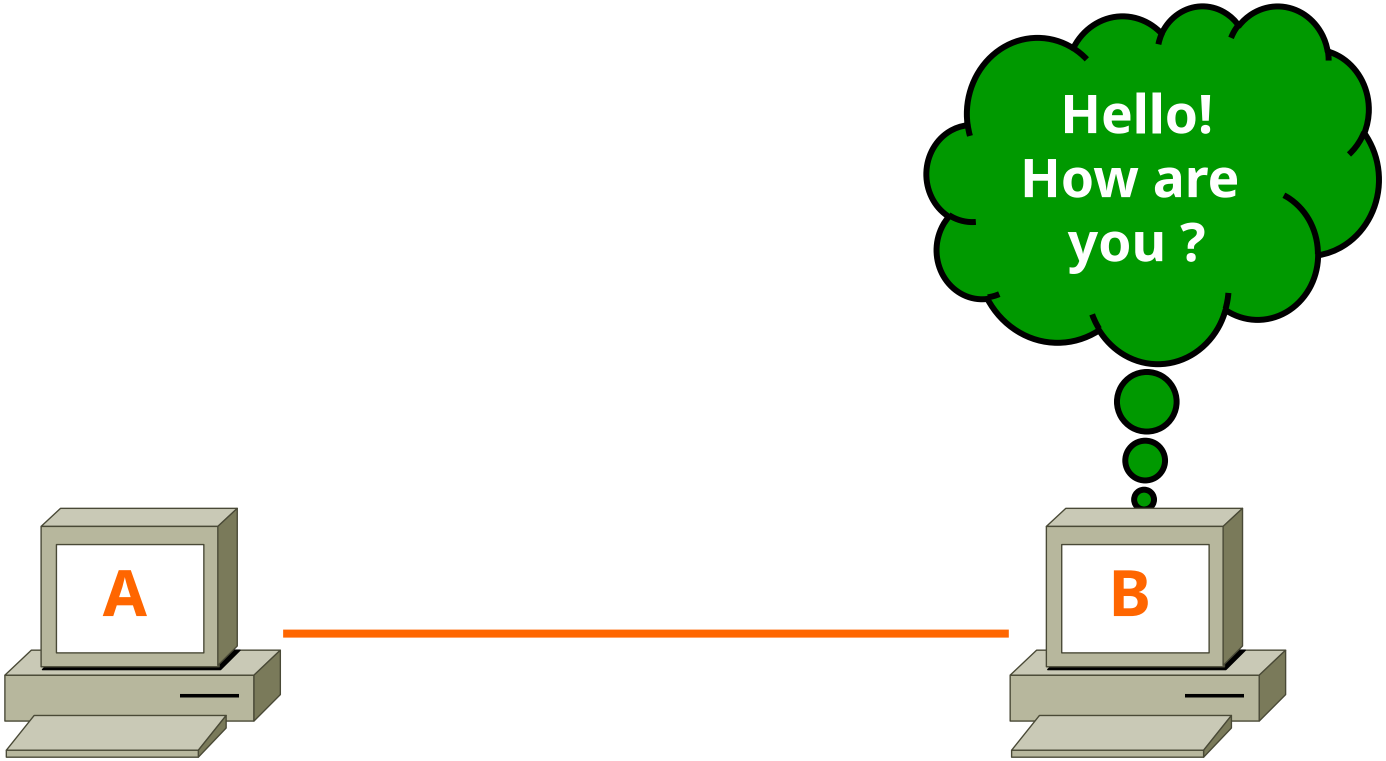
Segmentation



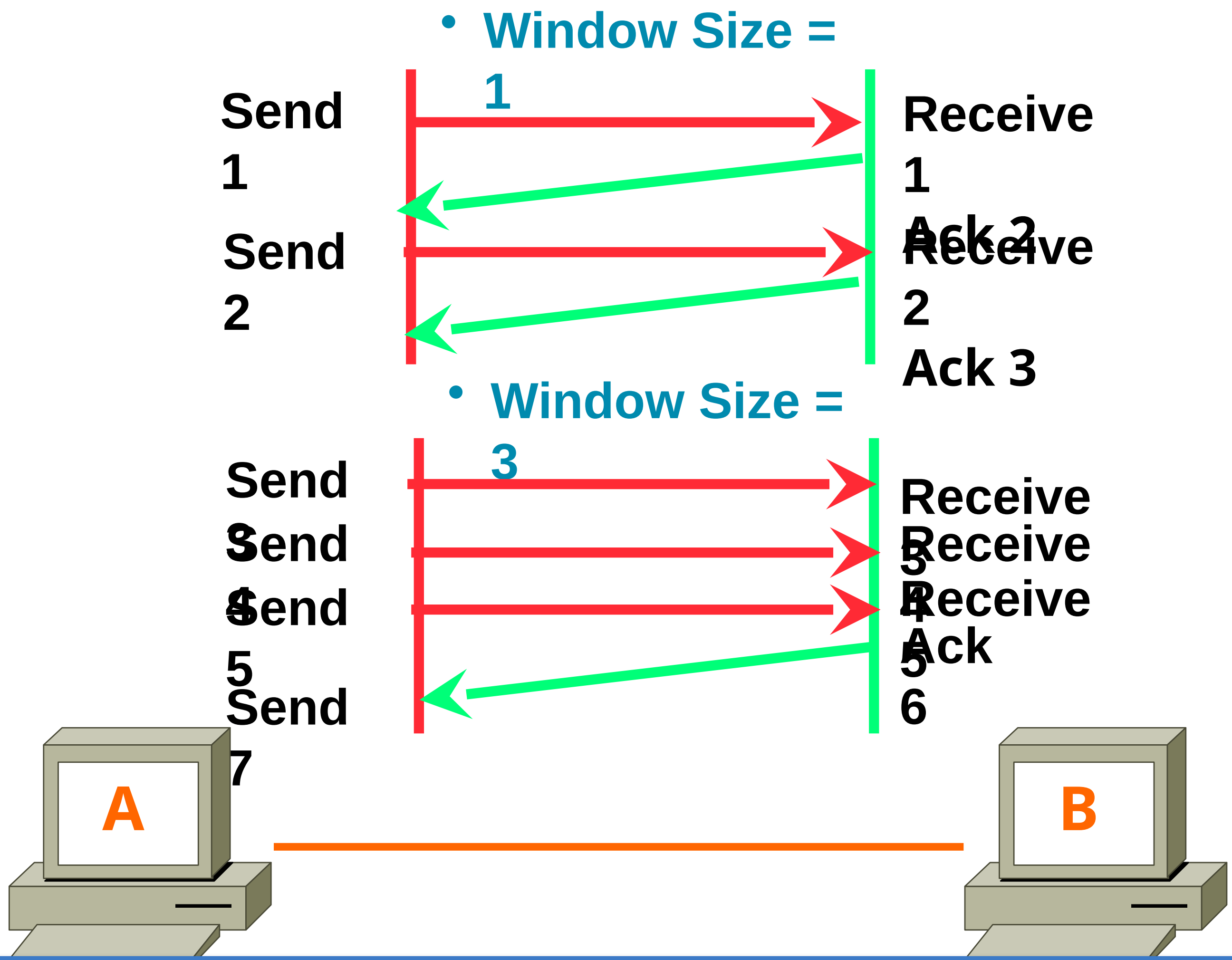
Sequencing & Reassembling

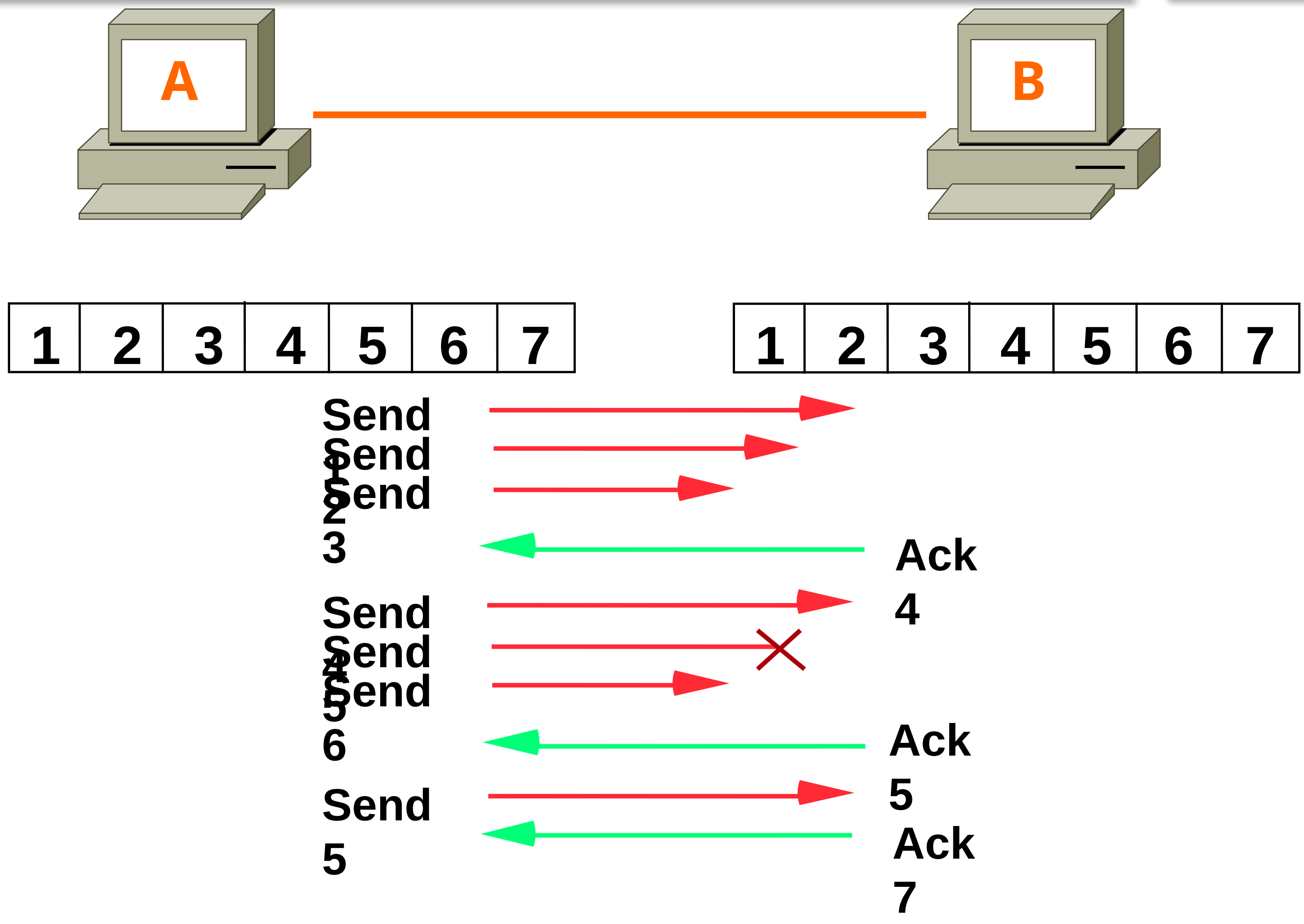


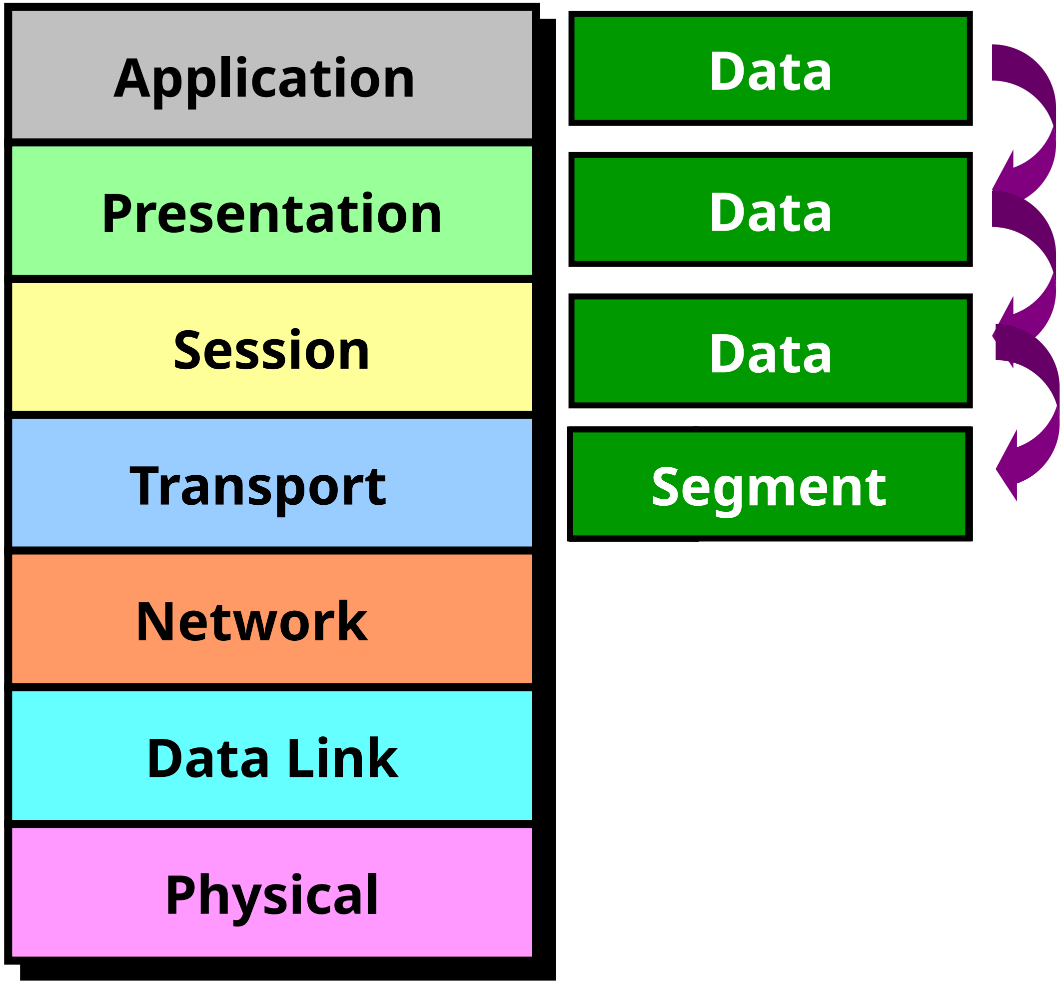
Error Correction



Flow Control




A
B
1 2 3 4 5 6 7
1 2 3 4 5 6 7
Send 1
Send 2
Send 3
Ack 4
Send 4
Send 5
Send 6
Ack 5
Send 5
Ack 7


Application
Presentation
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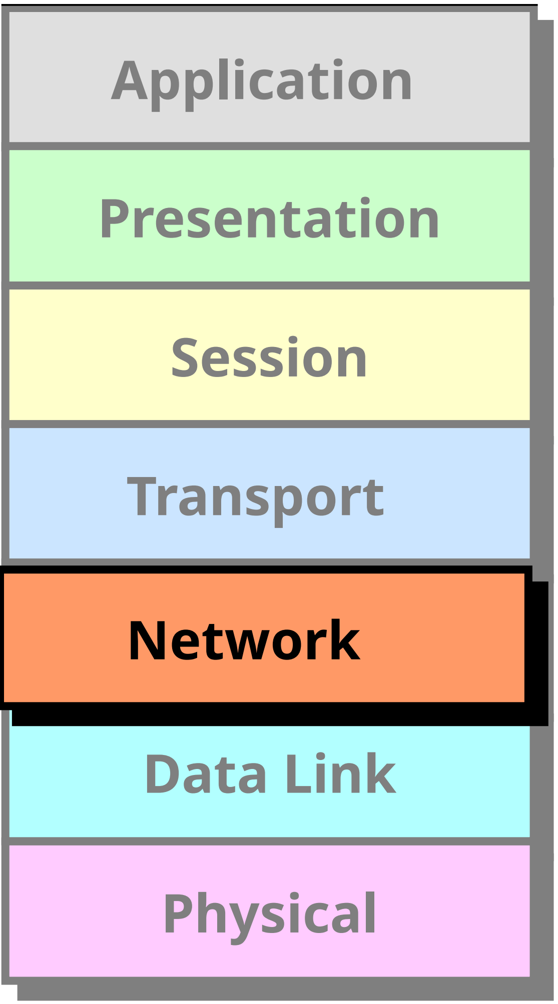
Data Link
Physical
Data
Data
Data
Segment



Network Layer

It provides Logical addressing & Path determination (Routing) in this layer.

The protocols that work in this layer are:

Routed Protocols:

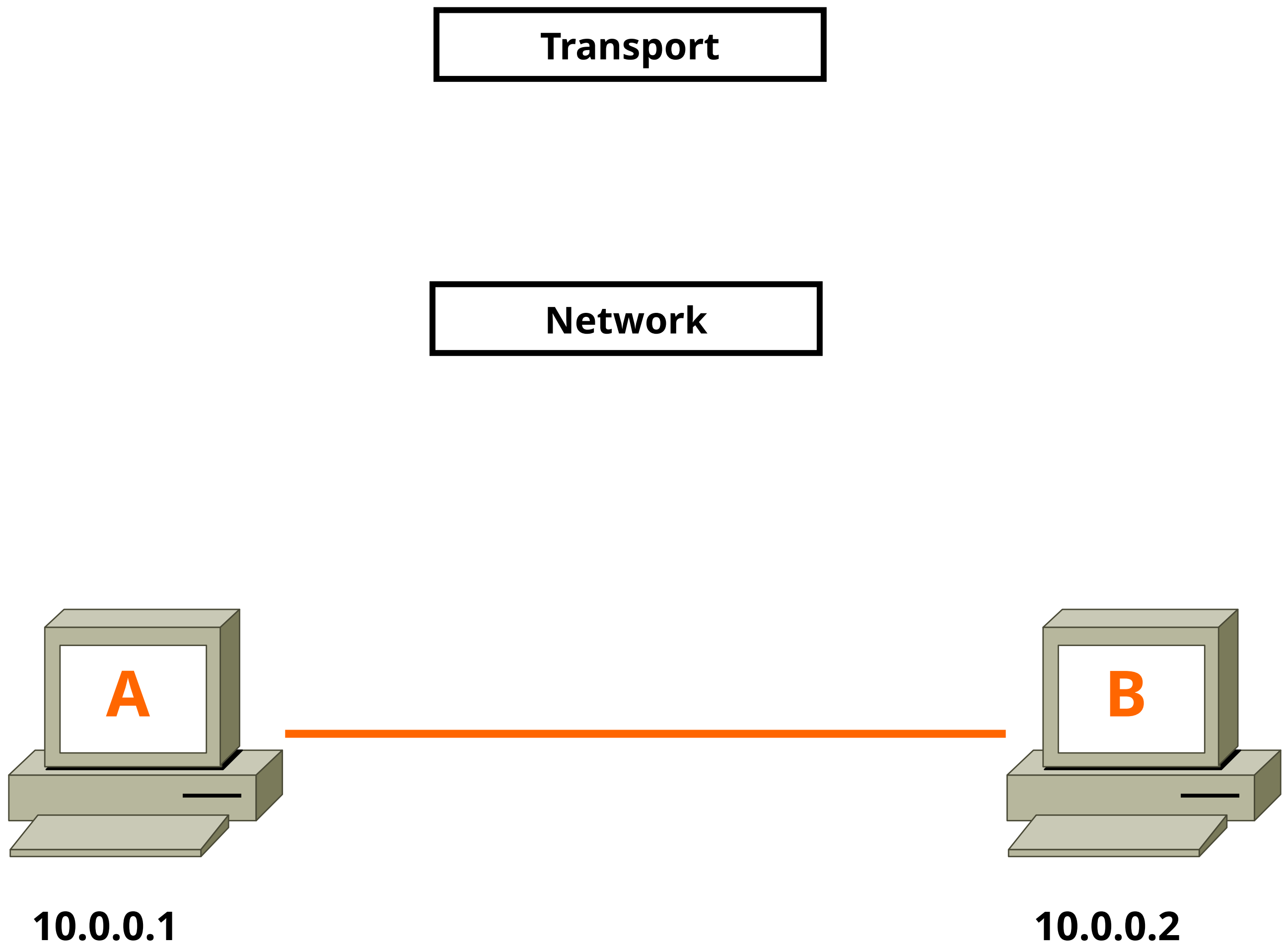
IP, IPX, AppleTalk.. Etc

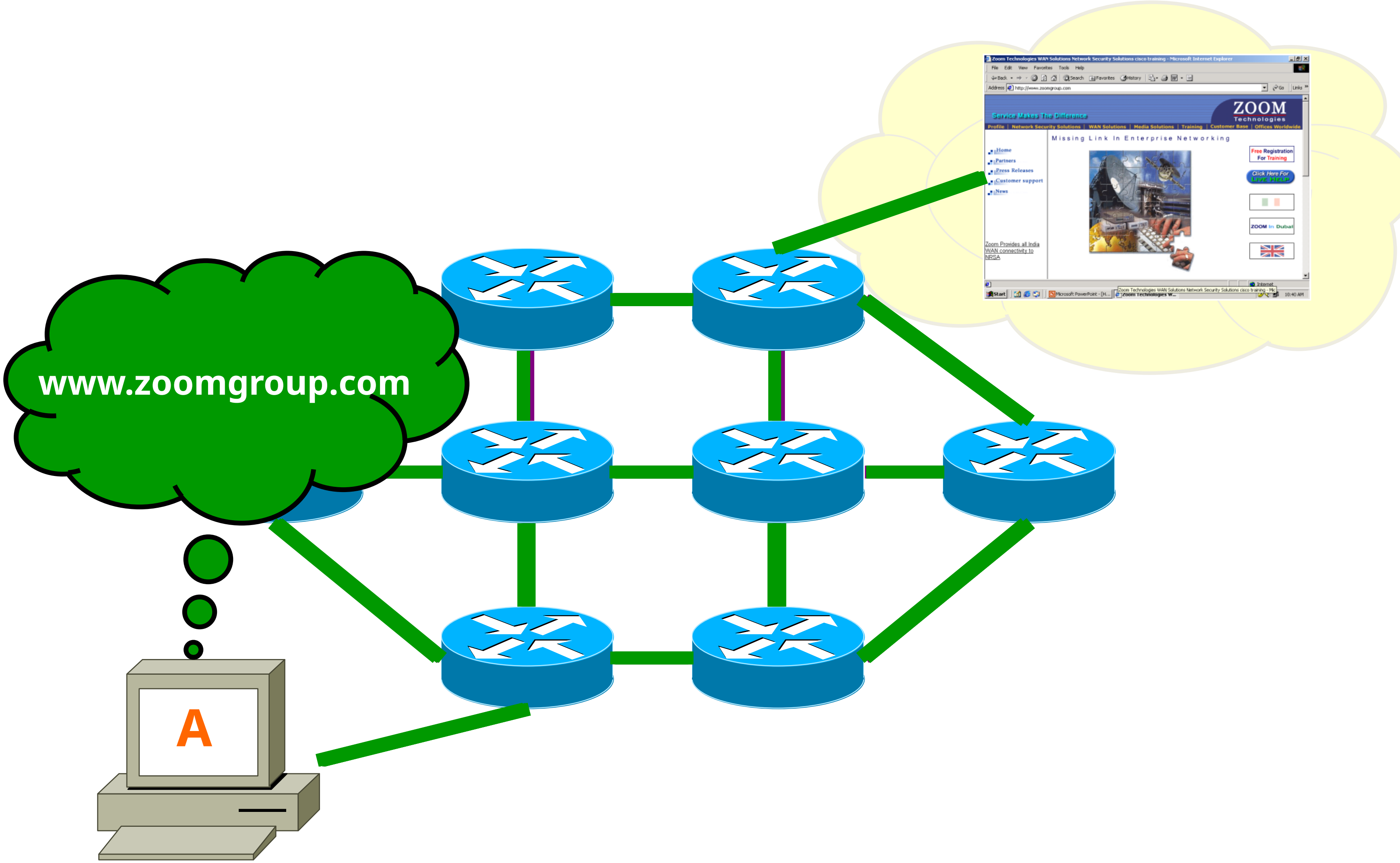
Routed protocols used to carry user data between hosts.

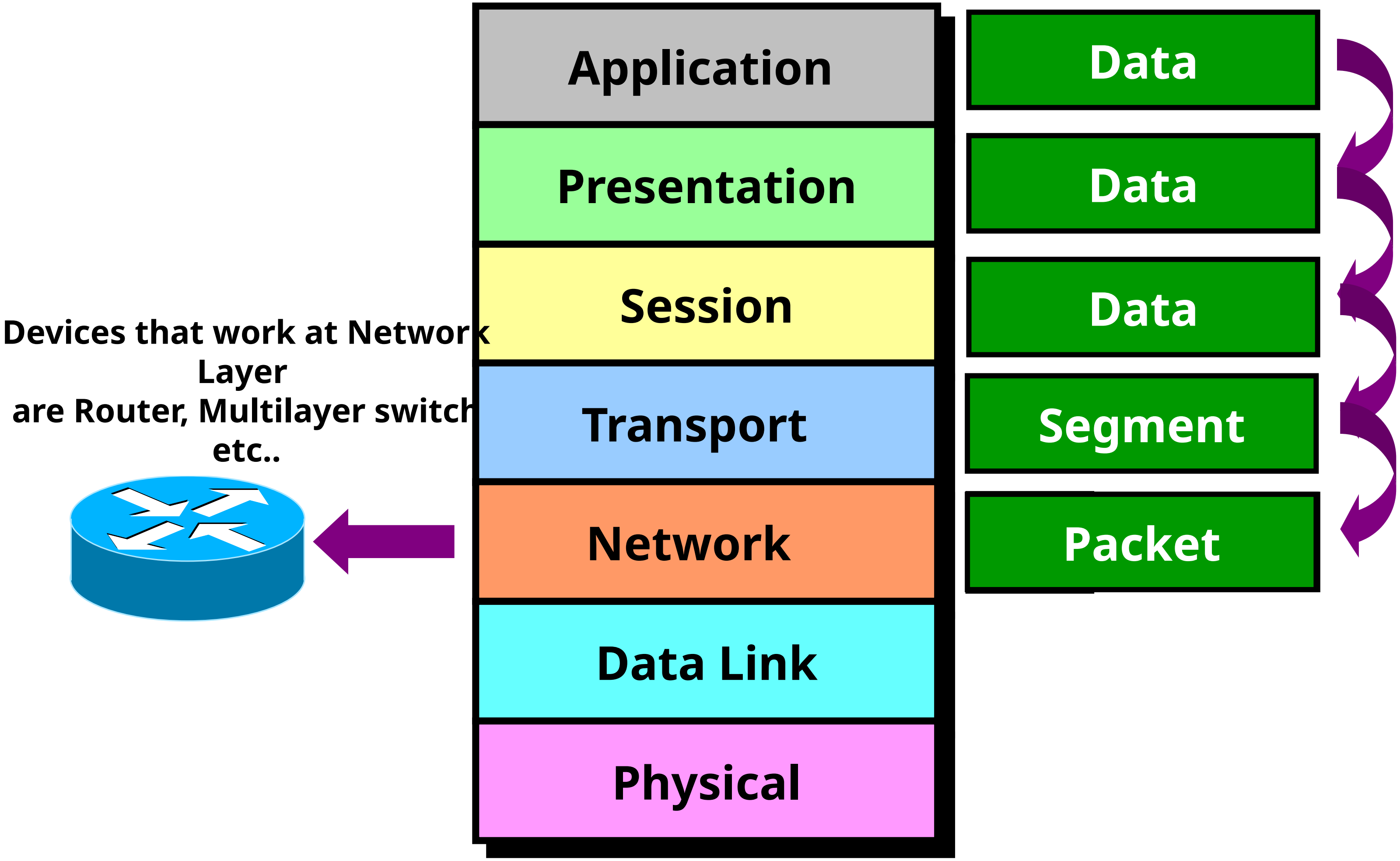
Routing Protocols:

RIP, OSPF.. Etc

Routing protocols performs Path determination (Routing).


Transport
Network
A
B
10.0.0.1
10.0.0.2


www.zoomgroup.com
A
ZOOM
Technologies
Missing Link In Enterprise Networking


Application
Presentation
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Network
Data Link
Physical
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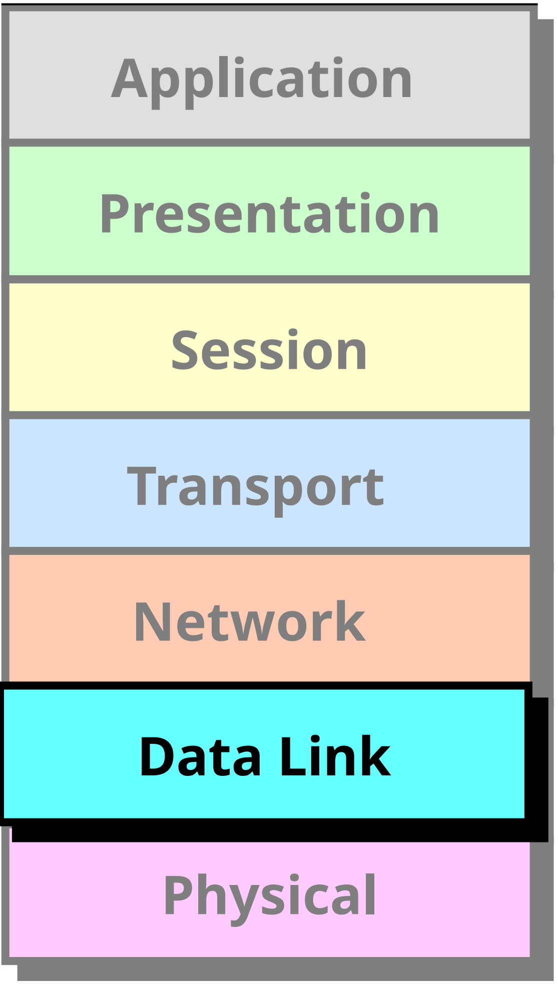
Data
Segment
Packet
Devices that work at Network Layer
are Router, Multilayer switch
etc..



Datalink Layer

It has 2 sub layers

- MAC (Media Access Control) It provides reliable transit of data across a physical link.

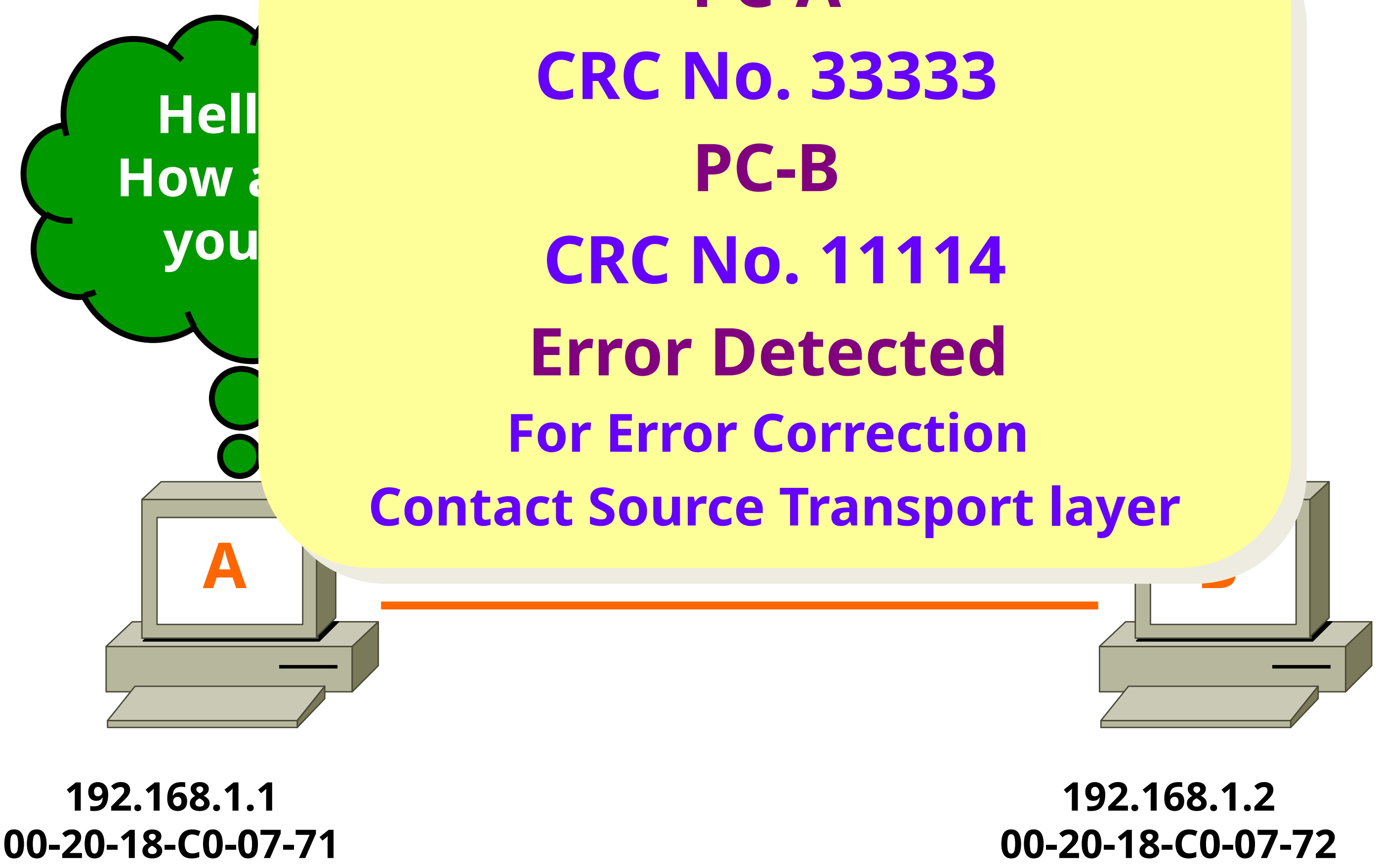
  It also provides ERROR DETECTION using CRC (Cyclic Redundancy Check) and ordered delivery of Frames.

  Ex: Ethernet, Token ring...etc

- LLC (Logical Link Control)

  It provides communication with Network layer.

  Negotiates with Network Layer using SAP & SNAP protocols



PC-A

CRC No. 33333

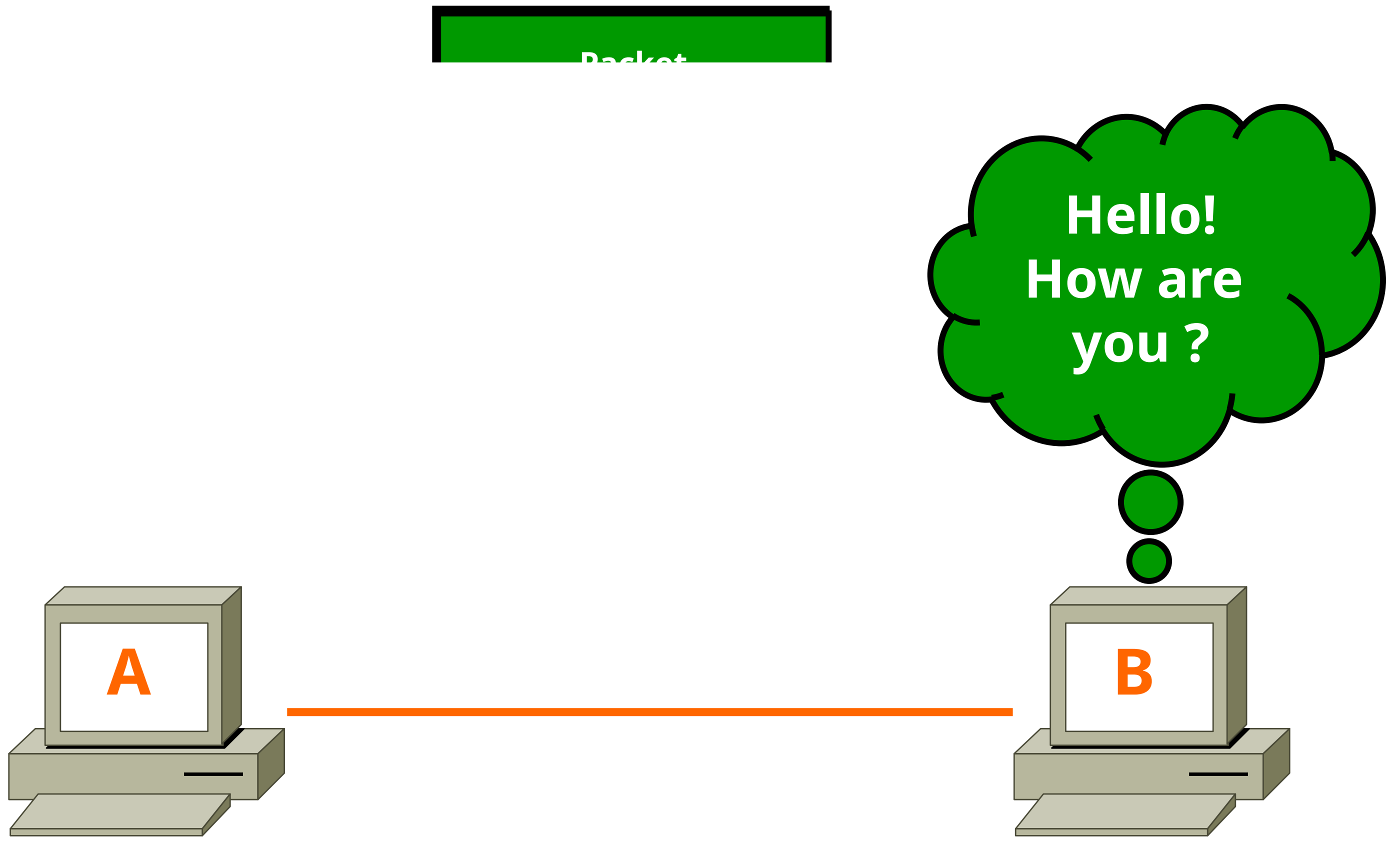
PC-B

CRC No. 11114

Error Detected

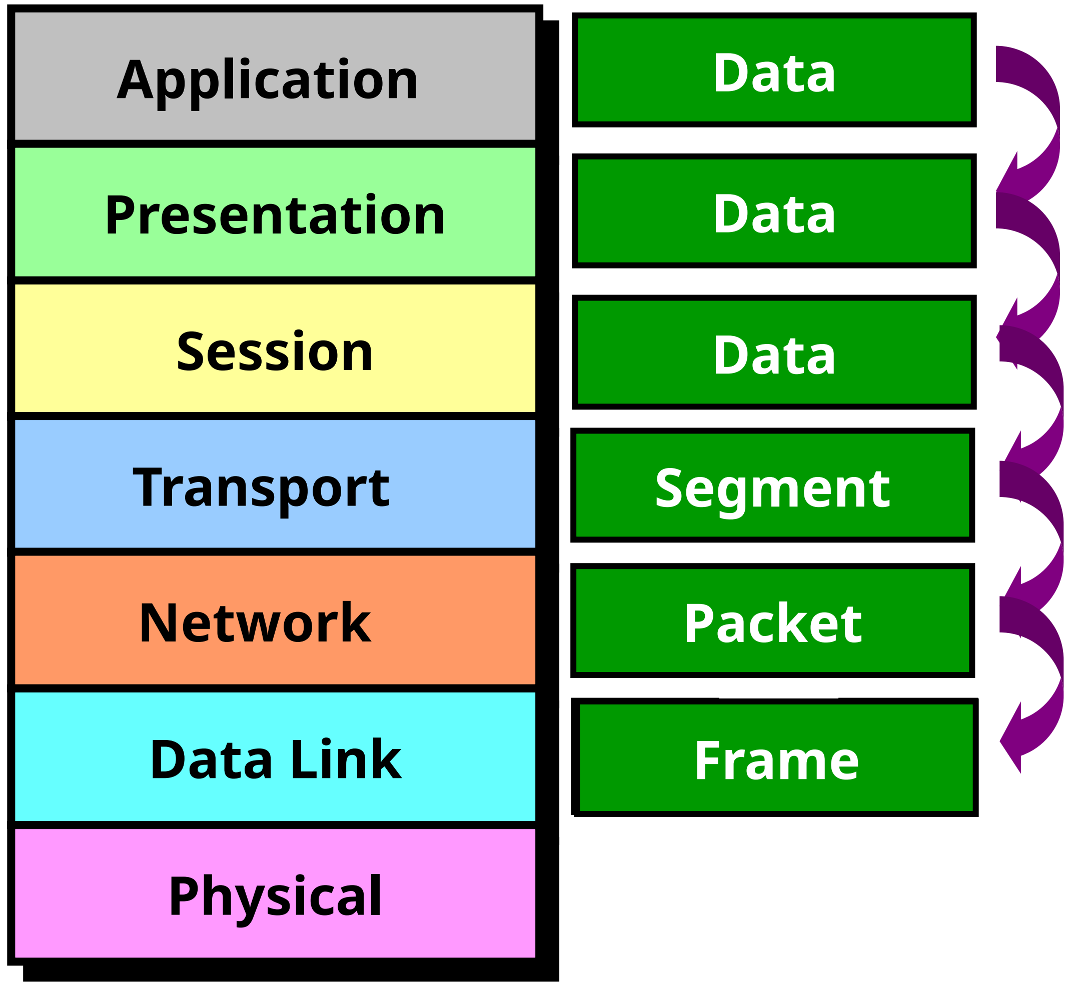
For Error Correction

Contact Source Transport layer

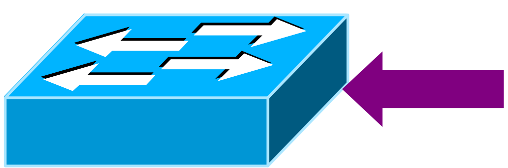


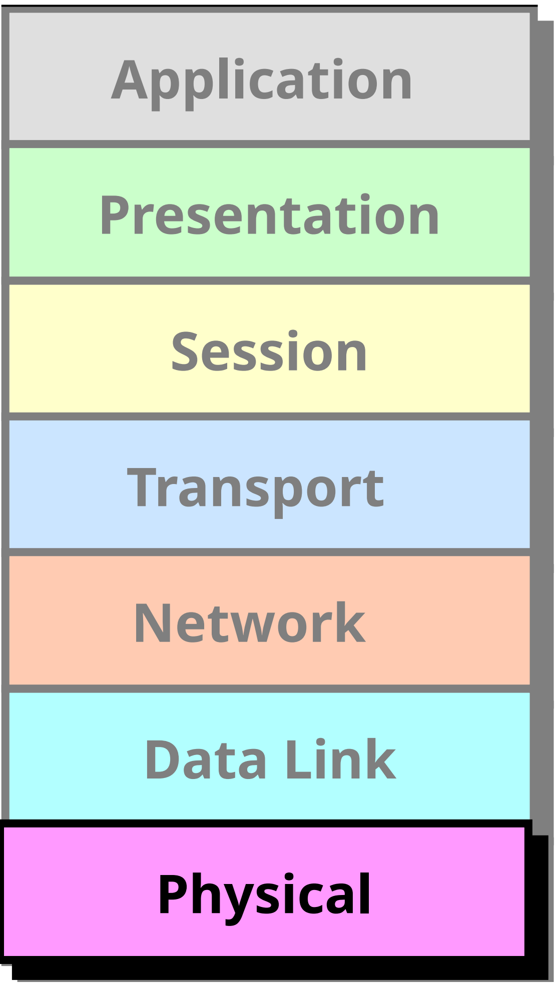
192.168.1.1
00-20-18-C0-07-71

192.168.1.2
00-20-18-C0-07-72



Devices that work at Data link layer are Switch, Bridge etc..





Physical Layer

It defines the electrical, Mechanical & functional specifications for communication between the Network devices.

The functions described at this layer are..

Encoding/decoding:

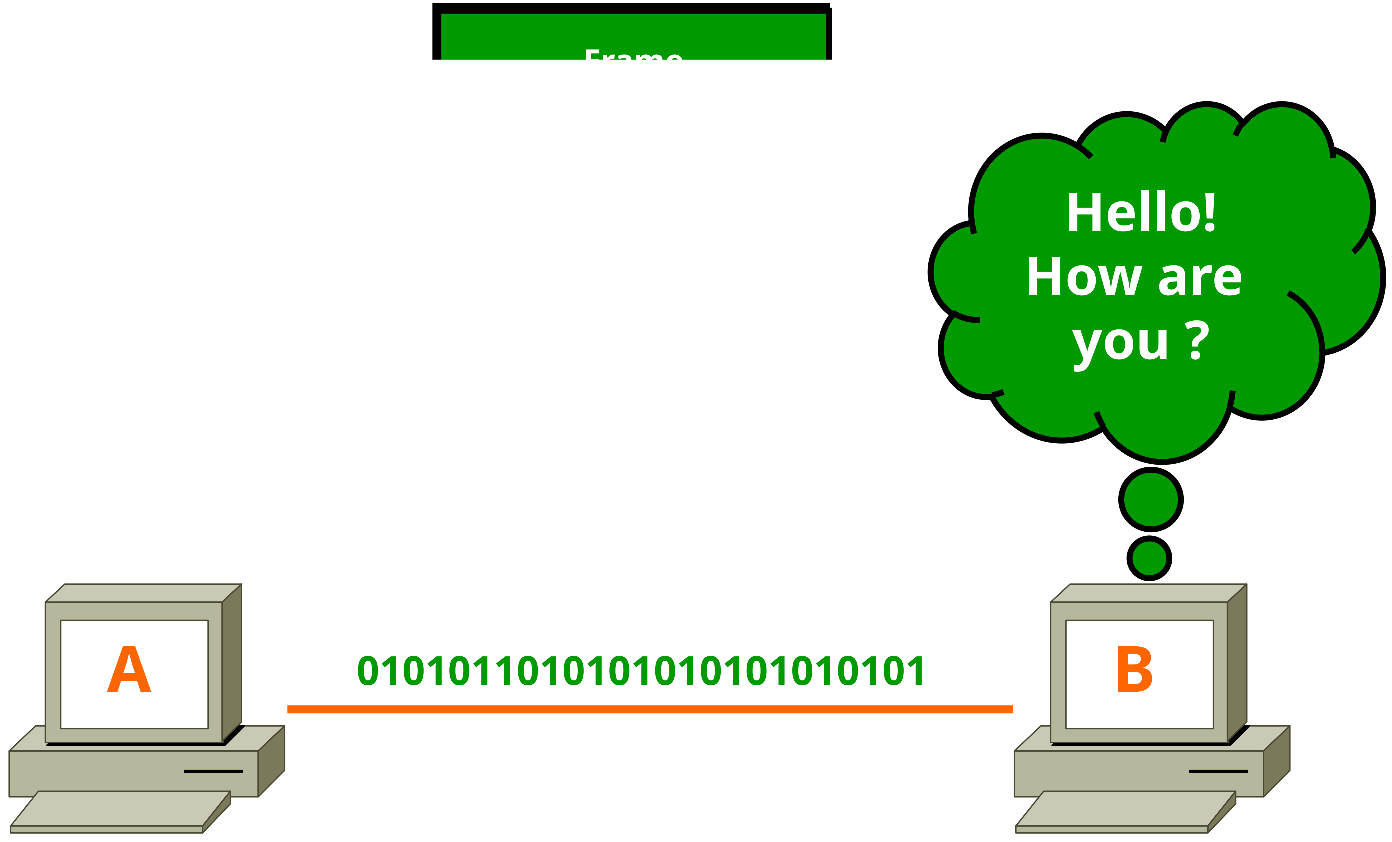
It is the process of converting the binary data into signals based on the type of the media.

- **Copper media** : Electrical signals of different voltages
- **Fiber media** : Light pulses of different wavelengths
- **Wireless media:** Radio frequency waves

Mode of transmision of signals:

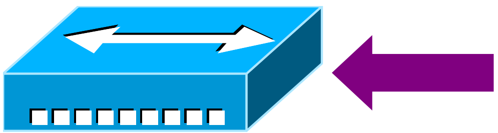
Signal Communication happens in three different modes
Simplex, Half-duplex, Full-duplex
Protocols works at physical layer: 10BaseT, 100BaseT,

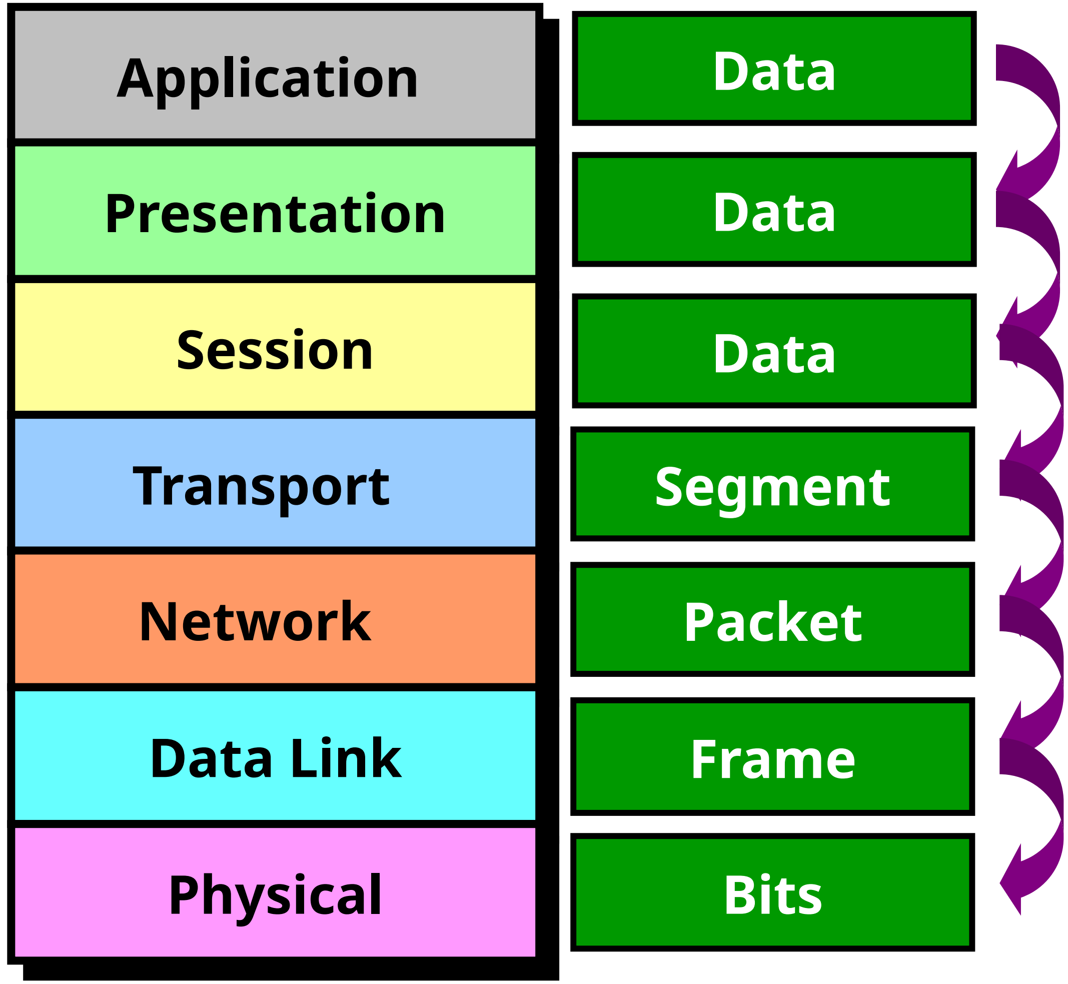
V.35, RS-232..etc

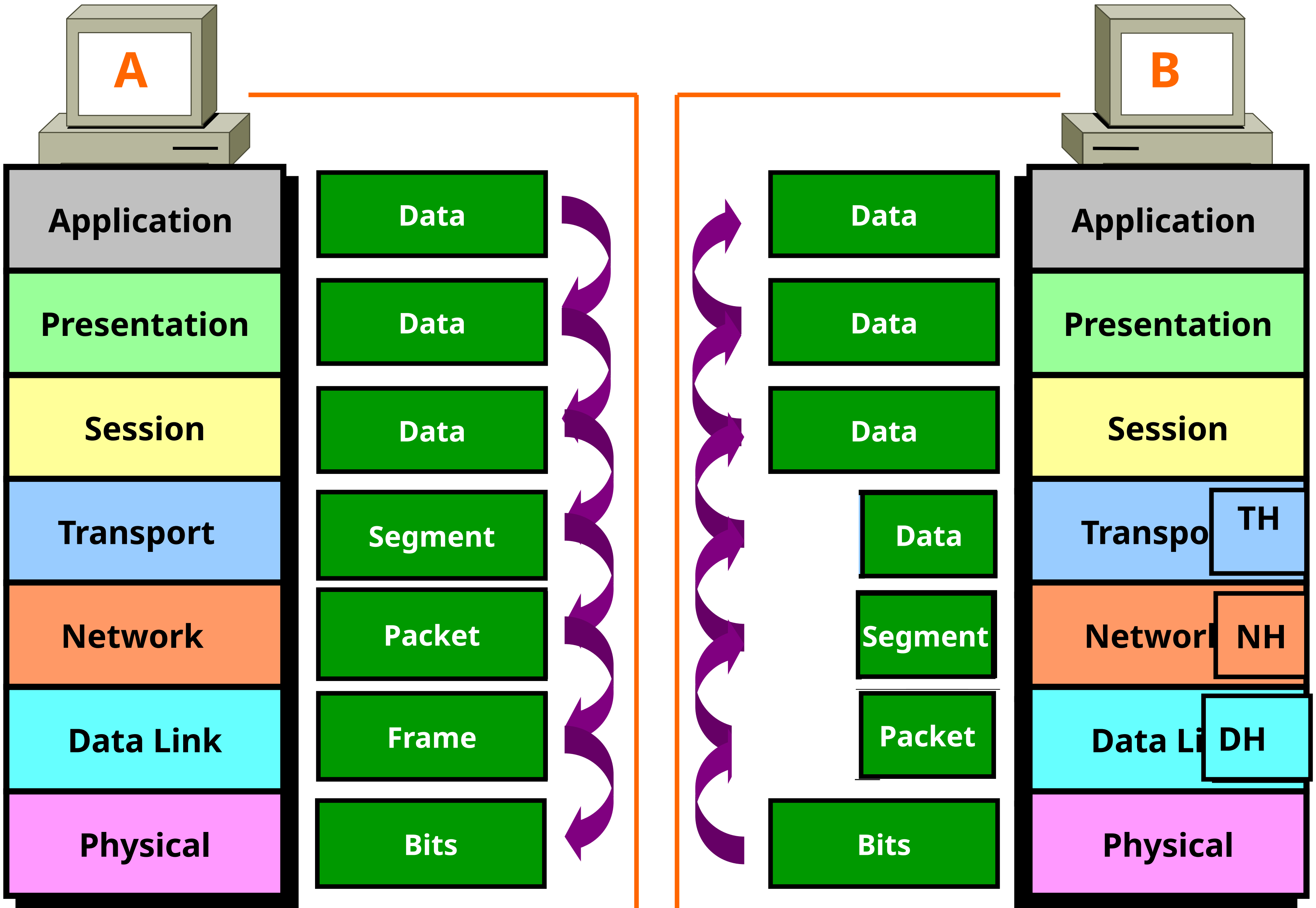


Hello!
How are you ?
A
0101011010101010101010101
B

Devices that work at physical layer are .. Hub, Repeater.. Etc






A
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Application
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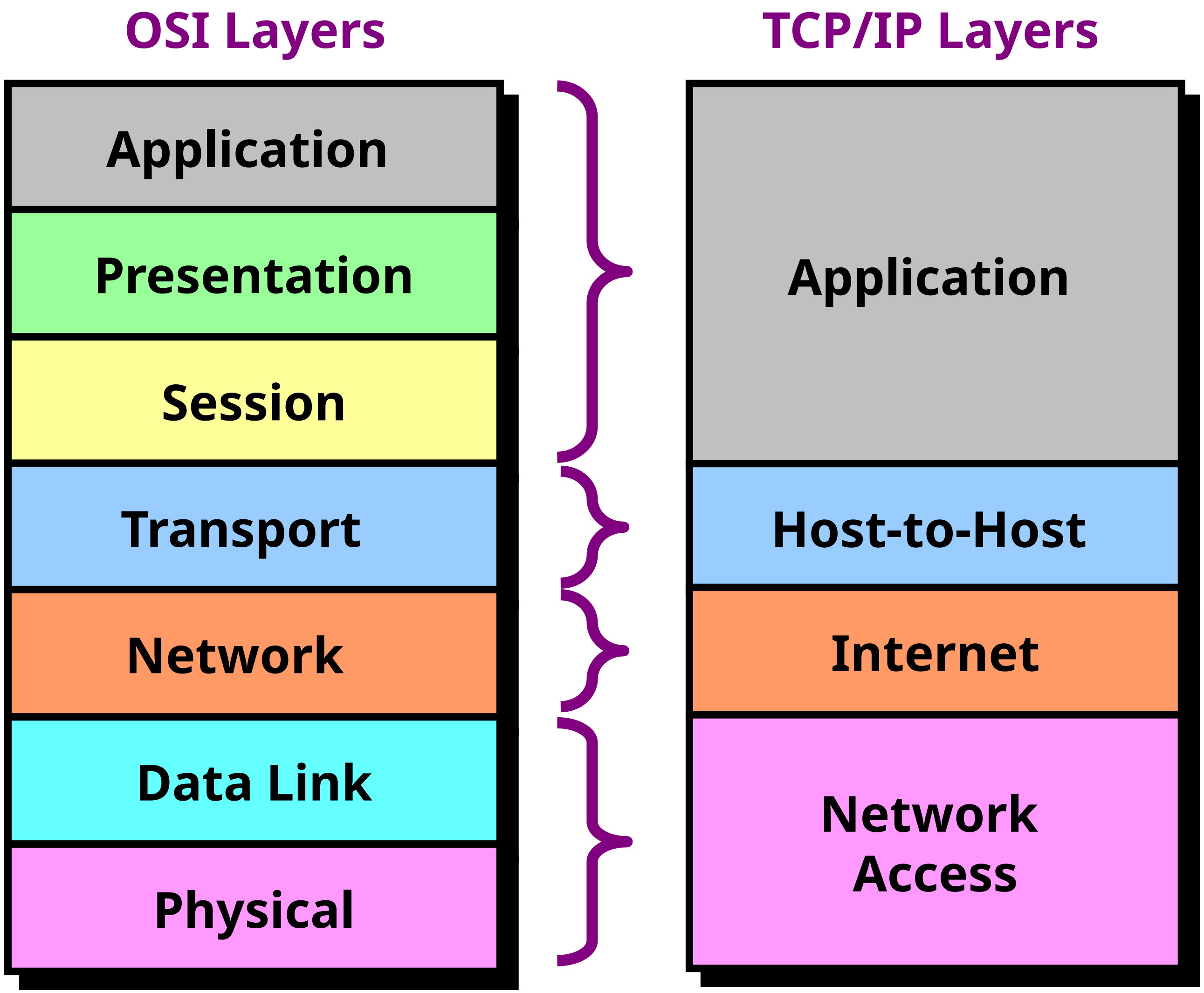
OSI Layers
Application
Presentation
Session
Transport
Network
Data Link
Physical
TCP/IP Layers
Application
Host-to-Host
Internet
Network
Access

THE END

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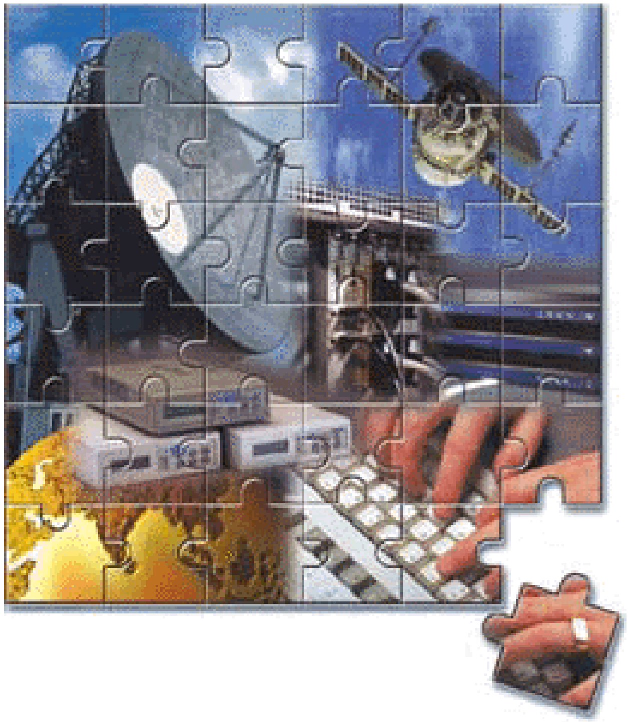

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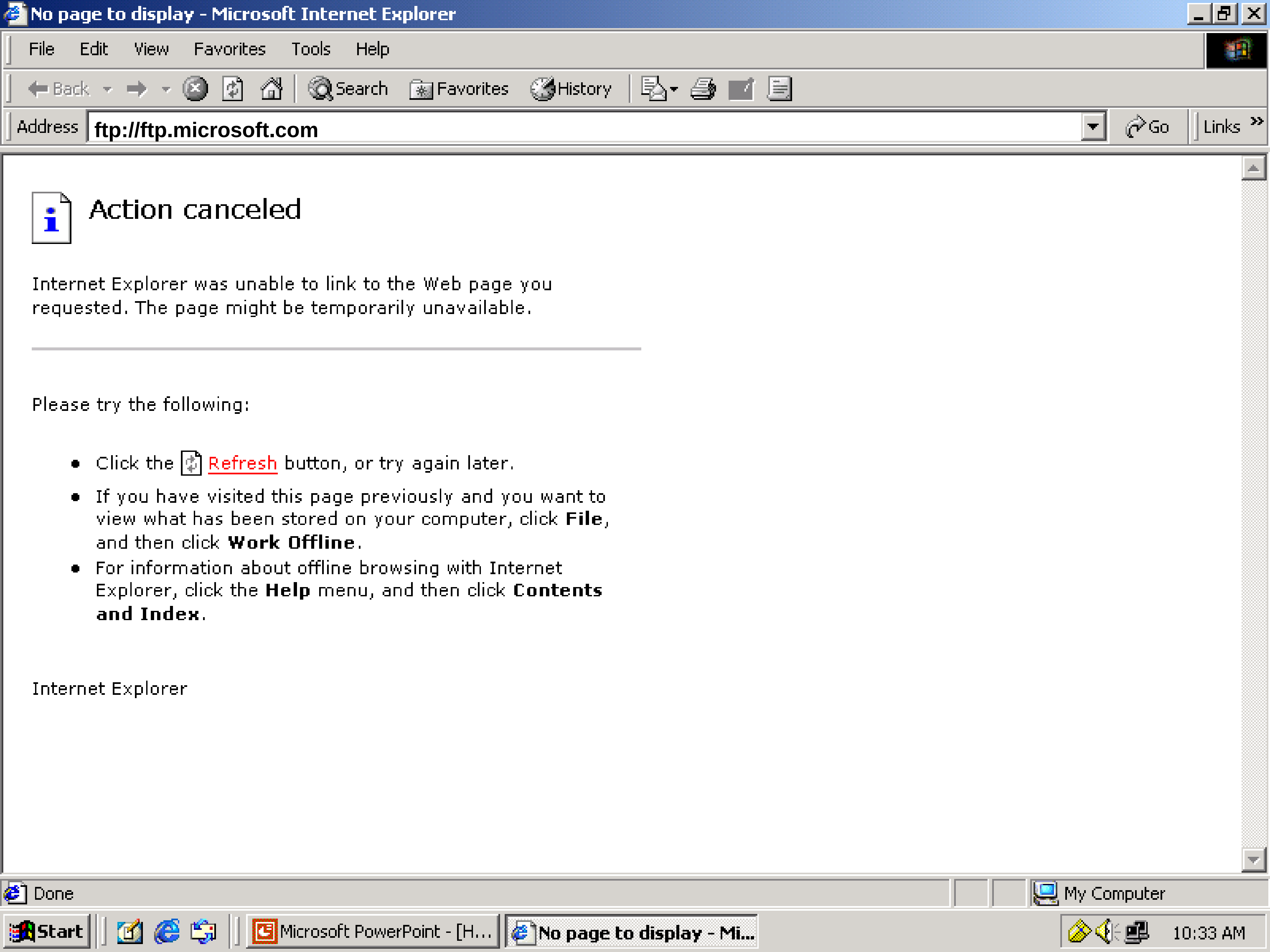
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Address ftp://ftp.microsoft.com
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Click the Refresh button, or try again later.
If you have visited this page previously and you want to view what has been stored on your computer, click File, and then click Work Offline.
For information about offline browsing with Internet Explorer, click the Help menu, and then click Contents and Index.
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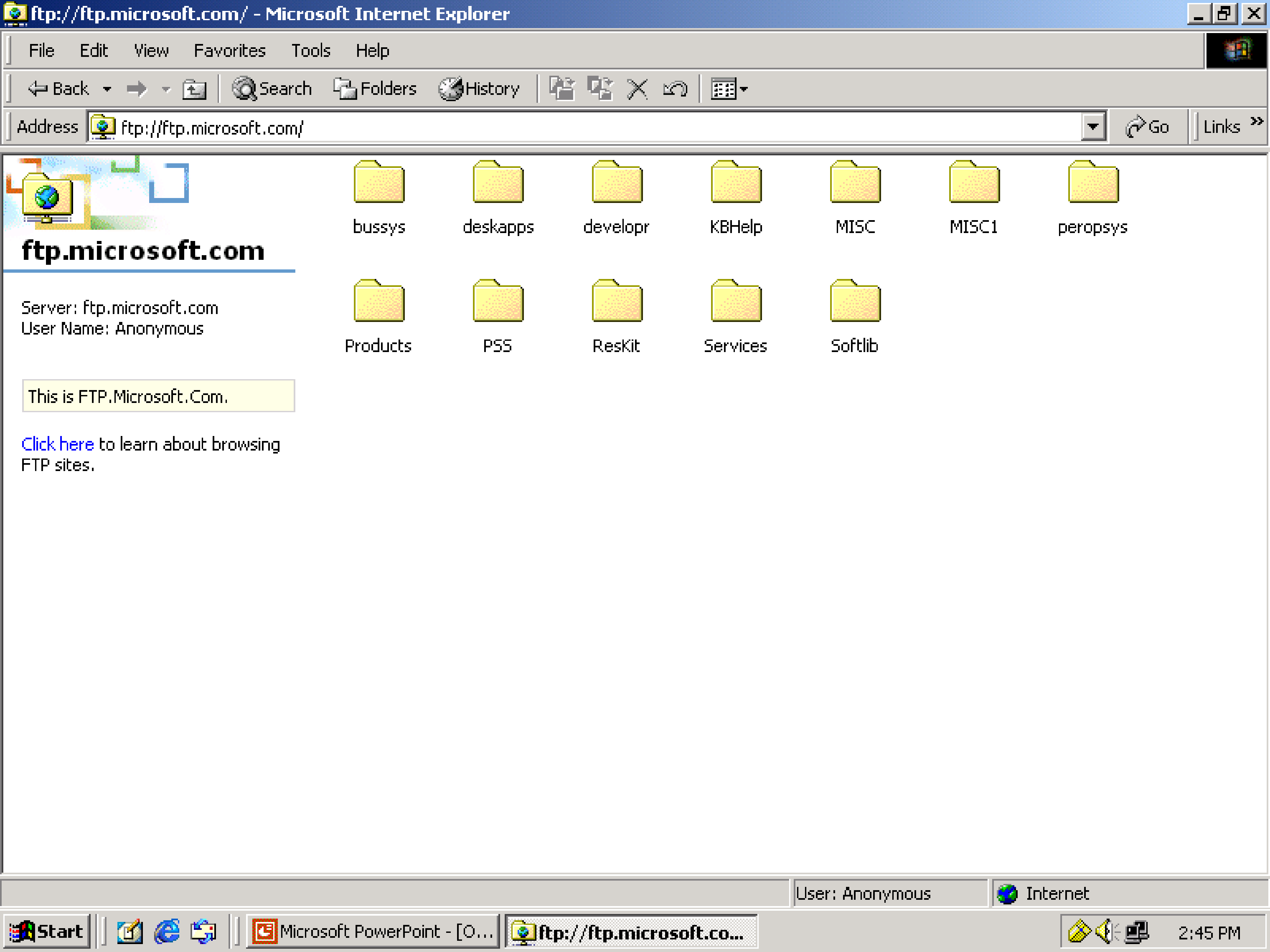
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User Name: Anonymous
This is FTP.Microsoft.Com.
Click here to learn about browsing FTP sites.
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deskapps
developr
KBHelp
MISC
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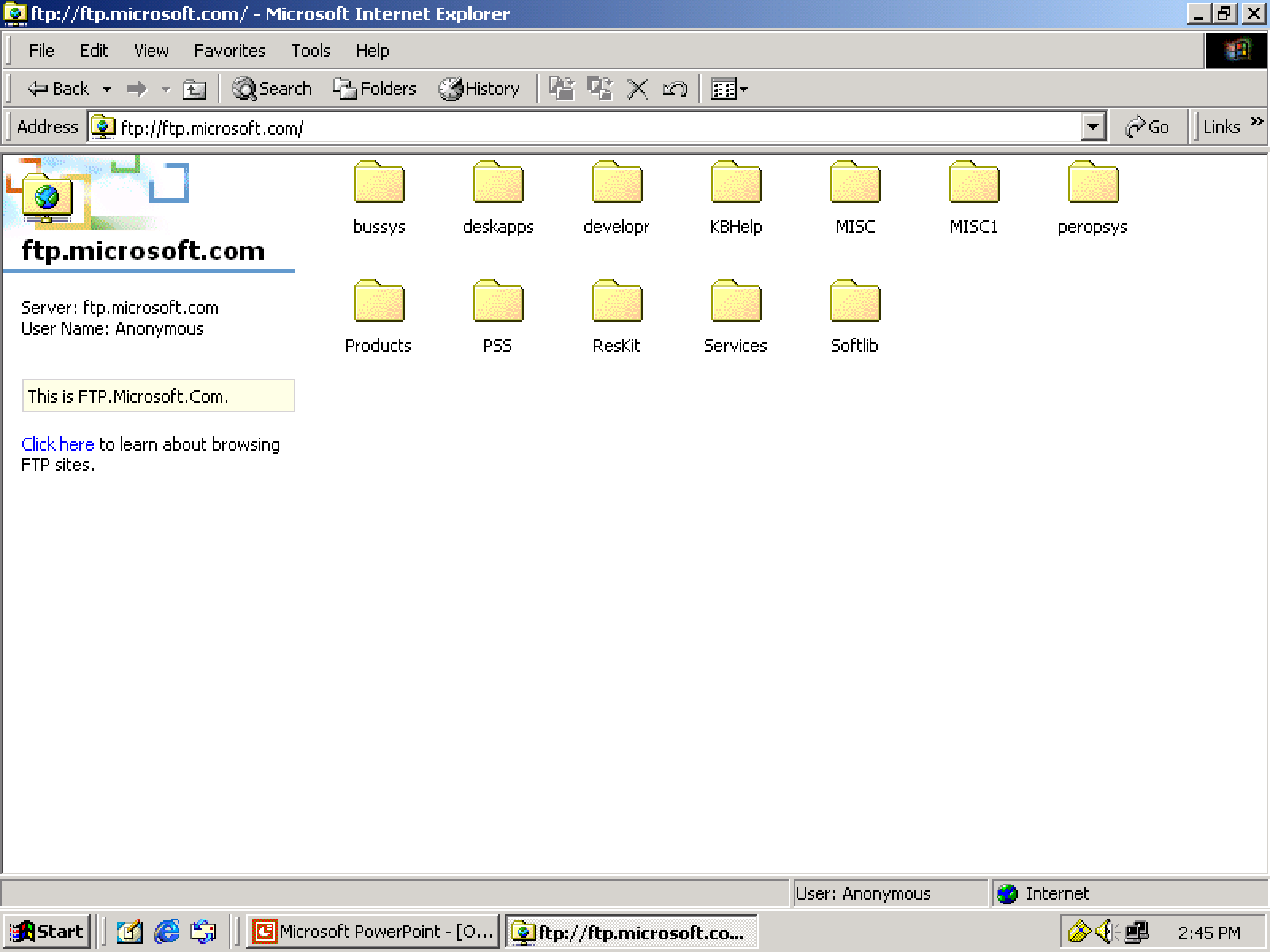
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ftp.microsoft.com
Server: ftp.microsoft.com
User Name: Anonymous
This is FTP.Microsoft.Com.
Click here to learn about browsing FTP sites.
bussys
deskapps
developr
KBHelp
MISC
MISC1
peropsys
Products
PSS
ResKit
Services
Softlib
User: Anonymous
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2:45 PM

C:\WINNT\system32\cmd.exe

```
Microsoft Windows 2000 [Version 5.00.2195]
(C) Copyright 1985-2000 Microsoft Corp.
C:\> telnet 192.168.1.150
Connecting .....
================================
Welcome to Hyderabad Router
================================
User Access Verification
password :
```

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Explore the Yahoo! Network

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Find	HotJobs, Maps, People Search, Personals, Yellow Pages
Connect	Chat, GeoCities, Greetings, Groups, Mail, Messenger, Mobile, Internet Access
Organize	Addresses, Briefcase, Calendar, My Yahoo!, PayDirect, Photos

Done Internet

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